Chapter 7 Matrix Representation of Graphs

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7.1 Matrix Representation of Graphs

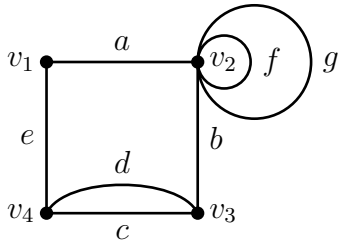
Clearly, we cannot represent a graph pictorially in a computer and hope to compute the parameters associated with graphs. More precisely, pictorial representations do not serve as data structures of graphs. We define below two convenient representations which can serve as data structures.

Definitions. Let G be a graph with vertices $v_1, v_2, \ldots, v_n$ and edges $e_1, e_2, \ldots, e_m$.
(1) The ***adjacency matrix*** $A(G) = [a_{ij}]$ of G, is the $n \times n$ matrix, where

$$a_{ij} = \begin{cases} \text{the number of edges joining } v_i \text{ and } v_j, & \text{if } i \neq j. \\ 2 \times (\text{the number of loops incident with } v_i), & \text{if } i = j. \end{cases}$$

Remarks.

- The adjacency matrix is a symmetric matrix.
- If G is simple, then every entry in $A(G)$ is zero or one and every diagonal entry is zero.
- Any matrix M in which every entry is zero or one is called a ***zero-one*** matrix or a ***binary*** matrix. So, $A(G)$ is a binary matrix if G is simple.



(a) G

	v_1	v_2	v_3	v_4
v_1	0	1	0	1
v_2	1	4	1	0
v_3	0	1	0	2
v_4	1	0	2	0

(b) $A(G)$

	a	b	c	d	e	f	g
v_1	1	0	0	0	1	0	0
v_2	1	1	0	0	0	2	2
v_3	0	1	1	1	0	0	0
v_4	0	0	1	1	1	0	0

(c) $B(G)$

Figure 7.1: A graph G with its adjacency matrix and incidence matrix.

(2) The ***incidence matrix*** $B(G) = [b_{ij}]$ of G, is the $n \times m$ matrix, where

$$b_{ij} = \begin{cases} 0, & \text{if } v_i \text{ is not incident with } e_j. \\ 1, & \text{if } v_i \text{ is incident with } e_j \text{ and } e_j \text{ is not a loop.} \\ 2, & \text{if } v_i \text{ is incident with } e_j \text{ and } e_j \text{ is a loop.} \end{cases}$$

Remark. If G is simple, then B(G) is a zero-one matrix.

Theorem 7.1. *Let $A = [a_{ij}]$ be the adjacency matrix of a simple graph G. Then the (i,j) th entry $[A^p]_{ij}$ in A^p is the number of walks of length p from v_i to v_j.*

Proof. We prove the theorem by induction on p. If $p = 1$, then A^p is A and the result is obvious. Suppose that the result is true for $p = r$ and let $p = r + 1$. We have,

$$[A^{r+1}]_{ij} = \sum_{k=1}^{n} [A^r]_{ik}\ a_{kj}.$$

Now

$$[A^r]_{ik}\ a_{kj} = \begin{cases} [A^r]_{ik}, & \text{if } a_{kj} = 1, \text{ that is, if } v_k \text{ and } v_j \text{ are adjacent.} \\ 0 & \text{if } a_{kj} = 0, \text{ that is, if } v_k \text{ and } v_j \text{ are not adjacent.} \end{cases}$$

By induction, $[A^r]_{ik}$ is the number of walks of length r connecting v_i and v_k. Each of these walks is extendable to a walk of length $r+1$ connecting v_i and v_j iff v_k and v_j are adjacent, that is, iff $a_{kj} = 1$. So, by the above equations, $[A^{r+1}]_{ij}$ is the number of walks of length $r+1$ connecting v_i and v_j. □

Theorem 7.2. *If G is a connected simple graph, then the distance between v_i and v_j is the smallest integer $p(\geq 0)$ such that $[A^p]_{ij} \neq 0$.*

Proof. By the minimality of p, $[A^r]_{ij} = 0$, for every $r, 0 \leq r \leq p-1$. So, by Theorem 7.1, there is no walk of length $\leq p-1$ connecting v_i and v_j; hence, $d(v_i, v_j) \geq p$. Since $[A^p]_{ij} \neq 0$, there does exist a walk, say $W(v_i, v_j)$, from v_i to v_j. Since, $[A^r]_{ij} = 0$, for all r, $0 \leq r \leq p-1$, $W(v_i, v_j)$ is a walk of minimum length and hence it is a path. So, $d(v_i, v_j) \leq p$. Combining the two inequalities we get $d(v_i, v_j) = p$. □

Rank of the Incidence Matrix

For an incidence matrix $B(G)$, the following observations follow immediately:

1. Since every edge is incident on exactly two vertices, each column of B has exactly two 1's.

2. The number of 1's in each row equals the degree of the corresponding vertex.

3. A row with all 0's, therefore, represents an isolated vertex.

4. Parallel edges in a graph produce identical columns in its incidence matrix.

5. If a graph G is disconnected and consists of two components, say G_1 and G_2, the incidence matrix $B(G)$ can be written in block-diagonal form:

$$B(G) = \begin{bmatrix} B(G_1) & 0 \\ 0 & B(G_2) \end{bmatrix}. \tag{7.1}$$

   The same observation is true for a disconnected graph with any number of components.

6. Permutation of any two rows or columns in an incidence matrix simply corresponds to relabeling the vertices and edges of the same graph.

Theorem 7.3. *If $B(G)$ is an incidence matrix of a connected graph G with n vertices, then $rank B(G) = n - 1$.*

Proof. Each row in an incidence matrix $B(G)$ may be regarded as a vector over $GF(2)$ in the vector space of the graph G. Let the vector in the first row be called B_1, in the second row B_2, and so on. Thus

$$B(G) = \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_n \end{bmatrix}.$$

Since there are exactly two 1's in every column of B, the sum of all these vectors is 0 (this being a modulo 2 sum of the corresponding entries). Thus the vectors $B_1, B_2, \ldots, B_n$ are not linearly independent. Therefore, $rank A < n$.

Now consider the sum of any m of these n vectors, where $m < n - 1$. If the graph is connected, $A(G)$ cannot be partitioned into block diagonal matrix as shown in Equation (7.1), with $B(G_1)$ containing m rows and $B(G_2)$ containing $n - m$ rows. In other words, no m by m submatrix of $B(G)$ can be found, for $m < n - 1$, such that the modulo 2 sum of those m rows is equal to zero.

Since there are only two constants, 0 and 1, in this field, the additions of all vectors taken m at a time for $m = 1, 2, \ldots, n - 1$ exhaust all possible linear combinations of $n - 1$ row vectors. Thus no linear combination of fewer than $n - 1$ row vectors of A can be equal to zero. Therefore the rank of $A(G)$ must be at least $n - 1$. □

Corollary. *If $B(G)$ is an incidence matrix of a simple graph G with n vertices, k components then $rank(B(G)) = n - k$.* □

If we remove any one row from the incidence matrix of a connected graph, the remaining $(n - 1) \times m$ submatrix is of rank $n - 1$. Such an $(n - 1) \times m$ submatrix $\mathbf{B_f}$ of B is called a ***reduced incidence matrix***. The vertex corresponding to the deleted row in B_f is called the *reference vertex*. Clearly, any vertex of a connected graph can be made the reference vertex.

Since a tree is a connected graph with n vertices and $n-1$ edges, its reduced incidence matrix is a square matrix of order $n-1$ and rank $n-1$. In other words:

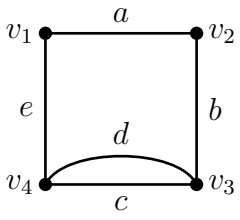
Corollary. *The reduced incidence matrix of a tree is nonsingular.* □

7.2 Circuit Matrix

Let the number of different circuits in a graph G be q and the number of edges in G be e. Then a circuit matrix $C_c(G) = [c_{ij}]$, of G is a $q \times m$ $(0,1)$-matrix defined as follows:

$$c_{ij} = \begin{cases} 1, & \text{if the } i\text{th circuit includes the } j\text{th edge,} \\ 0, & \text{otherwise.} \end{cases}$$

In Figure 7.2, the graph G has three different cycles or circuits, $\{a,b,c,e\}$, $\{a,b,d,e\}$, $\{c,d\}$. Thus, with edge order (a,b,c,d,e), we get the matrix as given in Figure 7.2b.



(a) G

$$C_c(G) = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

(b) Circuit Matrix

Figure 7.2: A graph G and its Circuit Matrix

The following observations can be made about a circuit matrix $B(G)$:

1. A column of all zeros corresponds to a noncircuit edge, that is, an edge that does not belong to any circuit.

2. Each row of $C_c(G)$ is a circuit vector.

3. Unlike the incidence matrix, a circuit matrix is capable of representing a self-loop; the corresponding row will have a single 1.

4. The number of 1's in a row is equal to the number of edges in the corresponding circuit.

5. If a graph G is separable (or disconnected) and consists of two blocks (or components) G_1 and G_2, the circuit matrix can be written in block-diagonal form:

$$C_c(G) = \begin{bmatrix} C_c(G_1) & 0 \\ 0 & C_c(G_2) \end{bmatrix}.$$

This follows because circuits in G_1 have no edges belonging to G_2, and vice versa.

6. Permutation of any two rows or columns in a circuit matrix simply corresponds to relabeling the circuits and edges.

Theorem 7.4. *Let C_c and B be, respectively, the circuit matrix and the incidence matrix of a self-loop-free graph, whose columns are arranged using the same order of edges. Then every row of C_c is orthogonal to every row of $B(G)$; that is,*

$$BC_c^T = C_c B^T = 0 \pmod 2.$$

Proof. Consider a vertex v and a circuit Γ in the graph G. Either v is in Γ or it is not. If v is not in Γ, there is no edge in the circuit Γ that is incident on v. On the other hand, if v is in Γ, the number of edges in the circuit Γ that are incident on v is exactly two.

Now consider the ith row in B and the jth row in C_c. Since the edges are arranged in the same order, nonzero entries in corresponding positions occur only if the particular edge is incident on the ith vertex and is also in the jth circuit.

If the ith vertex is not in the jth circuit, the dot product is zero. If the ith vertex is in the jth circuit, there are exactly two 1's in the sum of the products of corresponding entries. Since $1 + 1 = 0 \pmod 2$, the dot product is zero. Hence the theorem. □

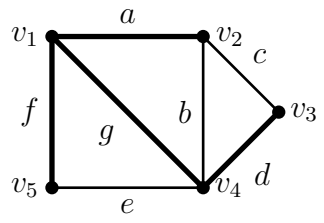
• Fundamental Circuit Matrix and Rank of C_c

Recall, that a set of fundamental circuits (or basic circuits) with respect to any spanning tree in a connected graph are the only independent circuits in a graph. The rest of the circuits can be obtained as ring sums, that is, linear combinations, of these circuits.

Thus, in a circuit matrix, if we retain only those rows that correspond to a set of fundamental circuits and remove all other rows, we do not lose any information. The remaining rows can be reconstituted from the rows corresponding to the fundamental circuits. For example, in the circuit matrix in 7.2, the third row is the modulo-2 sum of the first and second rows.

Definition. *A submatrix of a circuit matrix in which all rows correspond to a set of fundamental circuits is called a* ***fundamental circuit matrix*** C_c^f.

A graph G and its fundamental circuit matrix C_c^f is given in Figure 7.3. There are three fundamental circuits representing each of the chords b, c and e.



(a) G and its spanning tree

$$C_c(G) = \begin{array}{c} \\ C_1 \\ C_2 \\ C_3 \end{array} \begin{array}{c} \begin{array}{ccccccc} b & c & e & a & d & f & g \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} \end{array}$$

(b) Fundamental Circuit Matrix

Figure 7.3: A graph G and its Fundamental Circuit Matrix

If n is the number of vertices and m the number of edges in a connected graph, then C_c^f is an $(m-n+1) \times m$ matrix, because the number of fundamental circuits is $m-n+1$, each fundamental circuit being produced by one chord.

Let $\mu = m-n+1$ denote the nullity. Arrange the columns in C_c^f such that all the μ chords correspond to the first μ columns. Rearrange the rows so that the first row corresponds to the fundamental circuit made by the chord in the first column, the second row to the fundamental circuit made by the second chord, and so on.

A matrix C_c^f thus arranged can be written as

$$C_c^f = [\, I_\mu \mid C_c^t \,],$$

where I_μ is an identity matrix of order $\mu = m-n+1$, and C_c^t is the remaining $\mu \times (n-1)$ submatrix corresponding to the branches of the spanning tree.

It is clear that $rank(C_c^f) = \mu = m - n + 1$. Since C_c^f is a submatrix of the circuit matrix C_c, $rank(C_c) \geq m - n + 1$. In fact, we have the following theorem.

Theorem 7.5. *If C_c is a circuit matrix of a connected graph G with m edges and n vertices, then $rank(C_c) = m - n + 1$.*

Proof. If B is an incidence matrix of G, from Theorem 7.4, we have $BC_c^T = 0 \pmod 2$. Therefore, $rank(B) + rank(C_c) \leq m$. That is,

$$rank(C_c) \leq m - rank(B) \leq m - n + 1,$$

since $rank(B) = n-1$. But we already showed that $rank(C_c) \geq m-n+1$. Therefore, $rank(C_c) = m - n + 1$.

An alternative proof. Consider the circuit subspace W_Γ in the vector space W_G of a graph, as discussed in Unit 6. Every row in circuit matrix C is a vector in W_Γ. Hence $rank(C_c)$ = number of linearly independent rows in $B \leq \dim W_\Gamma$. Since $\dim W_\Gamma = m-n+1$, we again obtain $rank(C_c) \leq m-n+1$. Together with the lower bound already established, this proves the theorem. □

For a disconnected graph with k components, e edges, and n vertices, the corresponding component-by-component argument gives $rank(C_c) = m - n + k$.

7.3 Cut-Set Matrix

Analogous to a circuit matrix, we can define a cut-set matrix $C_s = [s_{ij}]$ in which the rows correspond to the cut-sets and the columns to the edges of the graph:

$$s_{ij} = \begin{cases} 1, & \text{if the } i\text{th cut-set contains the } j\text{th edge,} \\ 0, & \text{otherwise.} \end{cases}$$

The following remarks may be made about a cut-set matrix $C_s(G)$:

1. As in the case of the incidence matrix, a permutation of rows or columns in a cut-set matrix corresponds simply to renaming the cut-sets and edges, respectively.

2. Each row in $C_s(G)$ is a cut-set vector.

3. A column with all 0's corresponds to an edge forming a self-loop.

4. Parallel edges produce identical columns in the cut-set matrix.

5. In a non-separable graph, every set of edges incident on a vertex is a cut-set. Therefore every row of the incidence matrix $B(G)$ is included as a row in the cut-set matrix $C_s(G)$. In other words, for a non-separable graph G, $C(G)$ contains $B(G)$. For a separable graph, the incidence matrix of each block is contained in the cut-set matrix.

6. Hence $rank(C_s(G)) \geq rank(B(G))$. For a connected graph with n vertices, $rank C_s(G) \geq n-1$.

7. Since the number of edges common to a cut-set and a circuit is always even, every row in C_s is orthogonal to every row in C_c, provided the edges in both matrices are arranged in the same order: $C_c C_s^T = C_s C_c^T = 0 \pmod 2$.

We get $rank(C_c) + rank(C_s) \leq m$. For a connected graph, $rank(C_c) = m - n + 1$, and hence $rank(C_s) \leq n-1$. Combining $rank(C_s) = n-1$.

Theorem 7.6. *The rank of the cut-set matrix $C_s(G)$ is equal to the rank of the incidence matrix $B(G)$, which equals the rank of the graph G.*

As in the case of the circuit matrix, the cut-set matrix generally has many redundant, or linearly dependent, rows. Therefore it is convenient to define a fundamental cut-set matrix.

A **fundamental cut-set matrix** C_s^f of a connected graph G with m edges and n vertices is an $(n-1) \times m$ submatrix of C_s such that the rows correspond to the set of fundamental cut-sets with respect to some spanning tree.

A fundamental cut-set matrix can be partitioned into two submatrices:

$$C_s^f = [\, C_s^c \mid I_{n-1} \,], \tag{7-9}$$

where the last $n-1$ columns forming the identity matrix correspond to the branches of the spanning tree, and the first $m-n+1$ columns forming C_s^c correspond to the chords.

The discussion above has been stated for connected graphs. It can be generalized to disconnected graphs by considering one component at a time.

7.4 Relationships among B_f, C_c^f, and C_s^f

We now explore the relationships among the reduced incidence matrix B_f, the fundamental circuit matrix C_c^f, and the fundamental cut-set matrix C_s^f of a connected graph.

We have

$$C_c^f = [\, I_\mu \mid C_c^t \,], \quad \text{and } C_s^f = [\, C_s^c \mid I_{n-1} \,],$$

where C_c^t denotes the submatrix corresponding to the branches of a spanning tree, and C_s^c denotes the submatrix corresponding to the chords.

Let the spanning tree in both the expressions be the same, and let the order of the edges in both equations be the same. Furthermore, in the reduced incidence matrix B_f, of size $(n-1) \times m$, let the edges (columns) be arranged in the same order as in C_c^f and C_s^f. Partition B_f into

$$B_f = [\, B_c \mid B_t \,],$$

where B_t consists of the $n-1$ columns corresponding to the branches of the spanning tree T, and B_c is the remaining submatrix corresponding to the $m-n+1$ chords.

Since the columns in B_f and C_c^f are arranged in the same order,

$$\begin{aligned} B_f (C_c^f)^T &= 0 \pmod 2. \\ B_c + B_t C_c^{tT} &= 0. \end{aligned}$$

Since B_t is nonsingular, its inverse B_t^{-1} exists. Pre multiplying both sides by B_t^{-1} gives

$$B_t^{-1} B_c = -(C_c^t)^T = (C_c^t)^T,$$

since in modulo-2 arithmetic $-1 = 1$,

Similarly, because the columns in C_c^f and C_s^f are arranged in the same order, $C_s^f(C_c^f)^T = 0$. That is, $C_s^c = -(C_c^t)^T$ over $GF(2)$. Further, $C_s^c = B_t^{-1}B_c$. Above discussion leads to three conclusions:

1. Given B or B_f, we can readily construct C_c^f and C_s^f, starting from an arbitrary spanning tree and its subgraph B_t in B_f.

2. Given either C_c^f or C_s^f, we can construct the other.

3. Given either C_c^f or C_s^f, B_f in general cannot be determined completely.

7.5 Kirchoff-matrix-tree theorem

Kirchoff (1847) developed the theory of trees to find the current in each branch and around each circuit in an electrical network; any text book on network analysis contains these details. In this context, he was interested in finding the spanning trees of a given graph (representing an electrical network). Figure 7.4 shows the 8 distinct spanning trees of a graph.

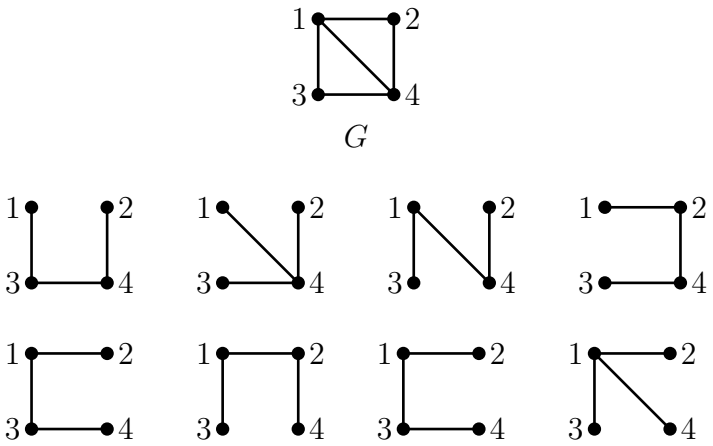


Figure 7.4: G and its eight distinct spanning trees

Remarks.

(1) The number of spanning trees of a graph G is denoted by $\mathcal{J}(G)$, where two spanning trees T_1 and T_2 are counted as distinct if $E(T_1) \neq E(T_2)$.
(2) $\mathcal{J}(G) = 0$ if and only if G is disconnected.
(3) $\mathcal{J}(G) = 1$ if and only if G is a tree.

(4) $\mathcal{J}(K_n) = \mathcal{J}_n = n^{n-2}$ (Cayley's formula).
(5) If $G^{(k)}$ denotes the graph obtained by replacing every edge of a simple graph G with k parallel edges, then $\mathcal{J}(G^{(k)}) = k^{n-1}\mathcal{J}(G)$.
(6) If G_0 denotes the simple graph obtained from a graph G by deleting all the loops of G, then $\mathcal{J}(G_0) = \mathcal{J}(G)$. So, in finding $\mathcal{J}(G)$ one can ignore the loops.

Kirchoff discovered an amazing formula for $\mathcal{J}(G)$. Cayley's formula can be derived using Kirchoff's formula. Let $K(G)$ denote the $n \times n$ matrix obtained from $A(G)$ by applying the following two operations:

1. Change the sign of every off-diagonal entry from + to −.

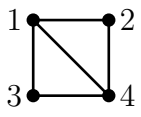
2. Replace the i-th diagonal entry by the degree of the i-th vertex.y

The matrix $K(G)$ is also called the **Laplacian** or **Kirchoff matrix** of G.

Theorem 7.7 (Kirchoff-Matrix-Tree-Theorem)**.** *Let G be a loopless graph, $A(G)$ and $K(G)$ be its adjacency and Laplacian matrices. Then all the $(n-1) \times (n-1)$-co-factors of $K(G)$ have the same value and the common value is $\mathcal{J}(G)$, that is $\mathcal{J}(G) = (-1)^{s+t} det(M_{(s,t)})$, where $M_{(s,t)}$ is the $(n-1) \times (n-1)$ matrix obtained from $K(G)$ by deleting the s-th row and the t-th column.*

Proof. Exercise. □

Illustration 1:



(a) G

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

(b) $A(G)$

$$\begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

(c) $K(G)$

$$\begin{bmatrix} 3 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 3 \end{bmatrix}$$

(d) $M(2,3)$

Figure 7.5: A graph G, its adjacency matrix, associated Kirchoff matrix and a co-factor.

$\mathcal{J}(G) = (-1)^{2+3} det(M(2,3)) = 8$. The eight distinct spanning trees of G are shown in Figure 7.4.

Illustration 2: Derivation of Cayley's formula using Kirchoff-Matrix-Tree Theorem.

$$\begin{bmatrix} 0 & 1 & 1 & \dots & 1 \\ 1 & 0 & 1 & \dots & 1 \\ 1 & 1 & 0 & \dots & 1 \\ & & \vdots & & \\ 1 & 1 & 1 & \dots & 0 \end{bmatrix}_{n\times n}$$

(a) $A(K_n)$

$$\begin{bmatrix} n-1 & -1 & -1 & \dots & -1 \\ -1 & n-1 & -1 & \dots & -1 \\ -1 & -1 & n-1 & \dots & -1 \\ & & \vdots & & \\ -1 & -1 & -1 & \dots & n-1 \end{bmatrix}_{n\times n}$$

(b) $K(K_n)$

$$\begin{bmatrix} n-1 & -1 & \dots & -1 \\ -1 & n-1 & \dots & -1 \\ & & \vdots & \\ -1 & -1 & \dots & n-1 \end{bmatrix}_{(n-1)\times(n-1)}$$

(c) $M(1,1)$

Apply row/column operations to $M(1,1)$ and reduce it to the following triangular matrix (for example, $R_1 \leftarrow R_1+R_2+\cdots+R_{n-1}$, $R_i \leftarrow R_1+R_i$, for $2 \leq i \leq n-1$).

$$\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & n & 0 & \dots & 0 \\ 0 & 0 & n & \dots & 0 \\ & & \vdots & & \\ 0 & 0 & 0 & \dots & n \end{bmatrix}_{(n-1)\times(n-1)}$$

$\mathcal{J}(K_n) = (-1)^2 det(M(1,1)) = n^{n-2}$.